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Petroleum Science and Technology

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/lpet20>

The Performance of a High Paraffin Reservoir Under Non-isothermal Waterflooding

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Published online: 20 Dec 2013.

To cite this article: O. D. Orodu & Z. Tang (2014) The Performance of a High Paraffin Reservoir Under Non-isothermal Waterflooding, Petroleum Science and Technology, 32:3, 324-334, DOI: [10.1080/10916466.2011.565295](https://doi.org/10.1080/10916466.2011.565295)

To link to this article: <http://dx.doi.org/10.1080/10916466.2011.565295>

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The Performance of a High Paraffin Reservoir Under Non-isothermal Waterflooding

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This study analyzes the performance of a high paraffin reservoir under cold waterflooding for 17 years using a 3-D finite difference simulator and analytical solution of injection wellbore temperature profile to upgrade reservoir management strategies. The reservoir has been marked by injectivity issues, early injection rate decline by half initial values and low incremental recovery, hence subject to alternate developmental schemes. The influence of cold waterflooding is assessed from simulated temperature maps due to nonisothermal injection and solution of injection well temperature profile.

Keywords: paraffin precipitation, pour point, non-isothermal waterflooding, reservoir simulation, Shen-95

INTRODUCTION

Block Shen-95 of Liaohe oilfield, characterized by high pour point oil (42–67°C, average of 51°C), paraffin content of 37.5%, and 12–20% resin and asphaltene content. The reservoir temperature is just a few degrees above the maximum pour point temperature. These factors make exploitation difficult and marked by low incremental recovery with regards to the peculiarity of formation damage at the vicinity of both production and injection wells. Formation damage is as a result of paraffin deposition from temperature decline at production wells and non-isothermal injection causing decreased production and low sweep efficiency. The reservoir is a heterogeneous sandstone reservoir with poor natural pressure support and has been on production for about 23 years with 17 years of cold waterflooding.

Early production problems attributed to the high pour point temperature had occurred and paraffin precipitation due to non-isothermal waterflooding was anticipated (Dou and Sun, 1995). The influence of non-isothermal flooding has been established by Ramey (1962), Platenkemp (1988), Civan (2000), Cassinat et al. (2002), and Maloney and Osthus (2005), all are related to paraffin crystallization, precipitation, entrainment, and deposition on dynamic and static reservoir properties. Typical investigations are restricted to wellbore and wellbore region. Analysis beyond the wellbore region is on waterflood sweep efficiency (Bedrikovetsky, 1997) and residual oil saturation effect on history matching (Mezzomo and Rabinovitz, 2001).

This study covers reservoir modeling, numerical simulation, and analytical solution of injection wellbore temperature profile. These provide the means of assessing non-isothermal waterflooding

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for high pour point crude oil reservoirs and its influence towards enhancing recovery by proffering optimization strategies. Optimization as regards to sustainable injection rates outside the cloud point and pour point temperature window, oil production rates and economics are looked at.

RESERVOIR DESCRIPTION AND MODEL

The use of different statistical approaches and property estimation enabled estimates of static and dynamic parameters that served as the basis for assigning these parameters to discretized cell blocks of the geological model by geostatistical tools. The ensuing reservoir model may not be a true representation of subsurface configuration leading to adjustment of model by history matching to capture reservoir performance.

Detailed reservoir characterization is as explained and carried out in Orodu et al. (2008) and geological setting in Orodu et al. (2009). The reservoir model was built on sequence stratigraphic layout of 17 layers. Porosity, permeability and water saturation were assigned by sequential Gaussian Simulation algorithm as part of the work flow of the reservoir modeling tool used. The fine grid was upscaled to $50\text{ m} \times 50\text{ m}$ ($110 \times 33 \times 17$ cells) for the North Fault Block and $50\text{ m} \times 50\text{ m}$ ($104 \times 64 \times 17$ cells) for the South Fault Block. Porosity and water saturation along well path were both averaged by arithmetic algorithm to cell size.

Faults were modeled from defined faults and flow characteristics of all faults initially assessed by production-injection wells interaction across faults. The analysis was based on significant early water breakthrough, wells placed in existing production-injection pair path; isolated production wells' early water breakthrough and inference of non-sealing/sealing fault from listed analysis. The analysis of dynamic production data as used was with regard to peak production/injection wells periods.

RESERVOIR SIMULATION

Simulation Model Initialization and Description

The reservoir model was initialized by incorporating recurrent data of observed oil, gas, and water production rates and water injection rates. Other data are fluid and reservoir properties (Table 1), equilibrium tables, physical property constants, kv/kh (sensitivity between 0.001 and 0.1), and the initial water saturation model. Rock, water, and crude oil thermal properties were also assigned. While the initial pressure equilibrium table consist of Shen-95 reservoir pressure at 1,800 m and that of the reservoir below 28.73 MPa at 2,300 m. This is a normal pressure gradient. Average reservoir pressure was not available but shut-in bottom-hole pressure (SIBHP) survey was available for a few wells.

A black oil simulator was used based on a dead oil model with the basic 3-D grid consisting of porosity, permeability, and NTG imported from the reservoir model.

Simulation History Matching

History matching was conducted separately for the North and South Fault Blocks with due consideration in minimizing simulation run time. Production-injection well pairs are absent across the fault boundary between both blocks and the fault was initially taken as a sealing fault. Similar criteria of production data, water breakthrough times, well (SIBHP) and original oil in place (OOIP) based

TABLE 1
Summary of Reservoir and Fluid Properties

Reservoir depth	1800–2230 m
Reservoir temperature	67.5°C
Average porosity	19%
Average permeability	3 mD
Net pay; North Block	12.2 m
Net pay South Block	20.26 m
Initial reservoir pressure @ 1800 m	17.98 MPa
Initial bubble point pressure	11.8 MPa
Initial oil saturation	66.3%
Initial GOR	70 m ³ /t
Initial oil boiling point	127–132°C
Oil formation volume factor	1.134
Oil density	0.8435–0.8991 g/cm ³
Pour point	42–64°C
Viscosity @ 100°C	3.83–9.67 mPa.sec
Cloud point	60°C
Paraffin content	37.5%
Colloid + asphaltene content	12–20%

on available data are used for evaluating match acceptability. The effect of paraffin precipitation is dealt with by sensitivity of endpoint saturation and relative permeability at the wellbore region and reduction of productivity index. This is as a result of the mechanics of paraffin precipitation, its entrainment, and deposition on production performance is rather complex.

North Fault Block

Block average pressure is not available for history match. Block watercut match was favorable except between 1998 and 2003 contributed to by unexplainable mismatch in two major and four minor wells with respect to their degree of influence (Figures 1a and 1b). These wells are close to injection wells, which account for 50% of producing wells as at that time. The match was run under oil rate control with 3 major relative permeability curves used for well match. A total of 23.7% of the wells had poor watercut match, and a well with SIBHP data showed reasonable and acceptable initial match. Breakthrough times for most wells are acceptable.

South Fault Block

Prior simulated average block pressure was extremely high at the third quarter of production period during peak water injection. Approaches at resolving this led to much higher watercut (Orudu et al., 2008). This study made use of flow-boundary on the major fault dividing the reservoir into North and South with significant reduced pressure in addition to modification of well flow parameters and the constrain on injection well bottom hole pressure. The situation may be likened to the study by Kasrale et al. (1993) of heavy-oil reservoir waterflooding. Simulation study had high pressure with the use of field injection rates but mitigated by adjusting injection rates under the guide of experiment confirmation of water migration into another subblock and NTG of 1 to duplicate field rates but with high OOIP. However, for this study, well injection rates were not adjusted and simulated block and well pressure are still rather high (Figures 2a and 2b).

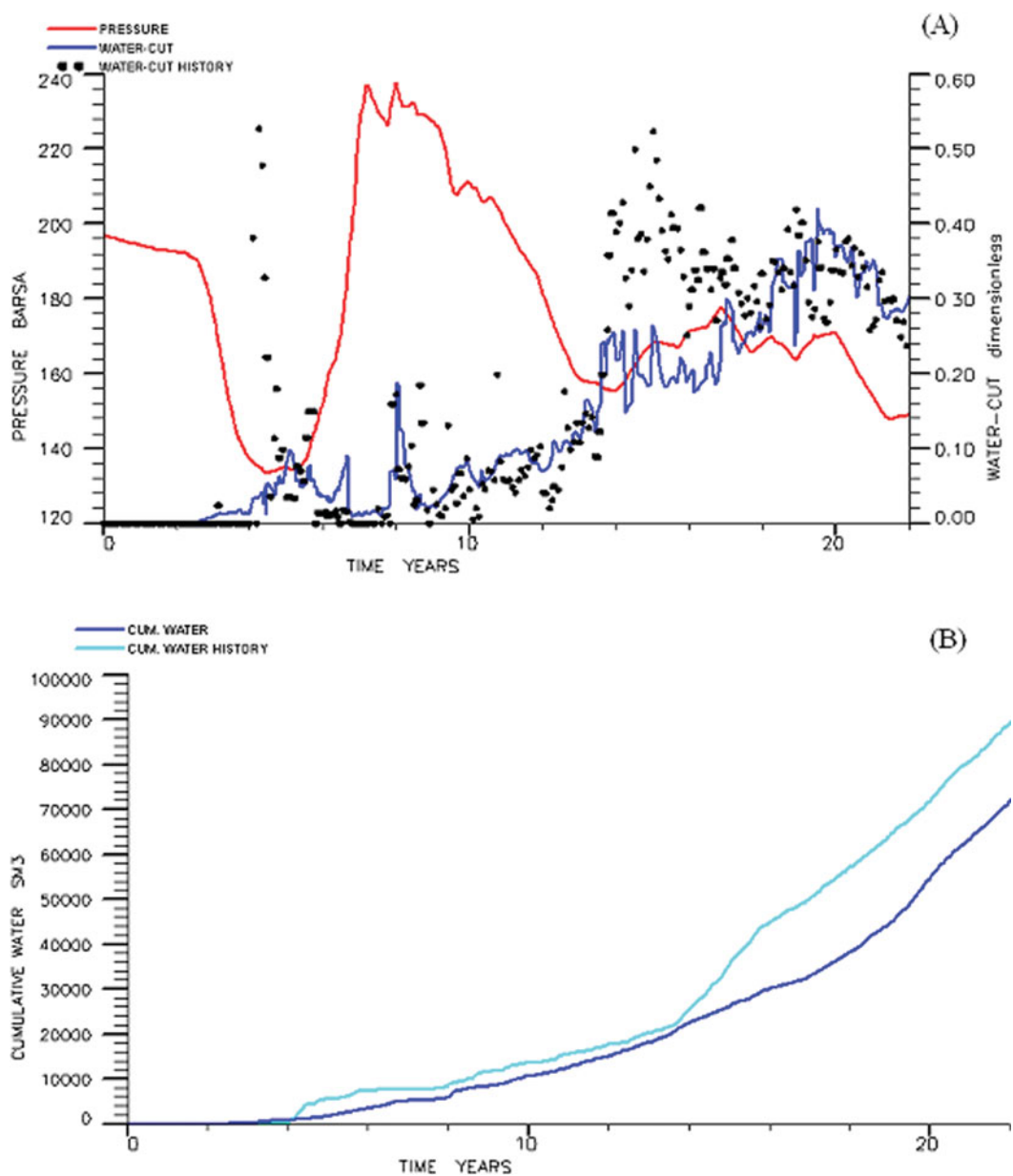


FIGURE 1 (A) North Fault Block watercut match and (B) North Fault Block cumulative produced water match.

RESERVOIR NON-ISOTHERMAL WATERFLOODING

Reservoir Temperature Distribution

Figures 3 and 4 shows the areal temperature distribution of a major productive layer at peak injection rates for both the North and South fault blocks. Viscosity is held constant with temperature as

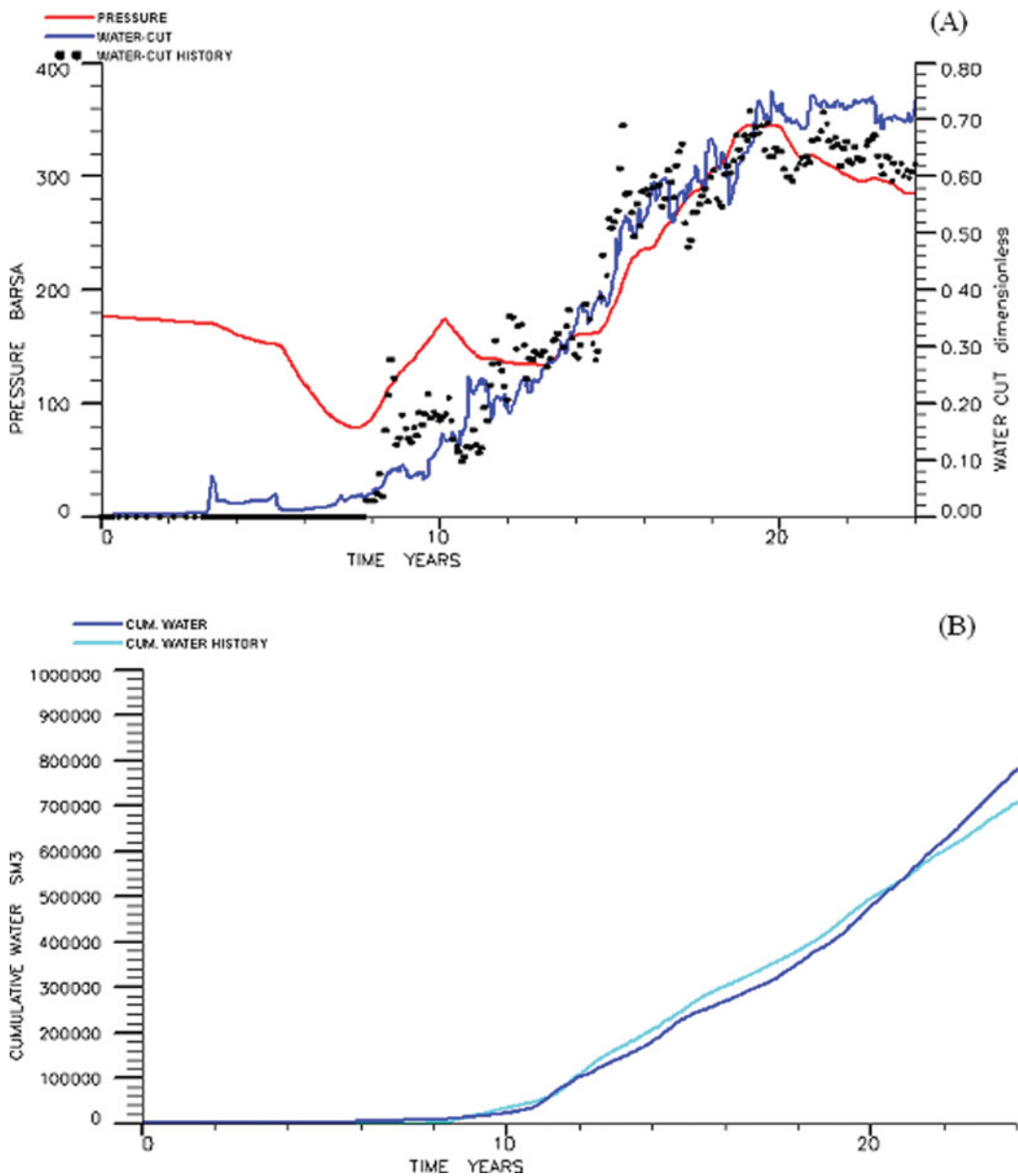


FIGURE 2 (A) South Fault Block watercut match and (B) South Fault Block cumulative produced water.

temperature change was initially assumed confined to near wellbore vicinity. Injection water temperature was varied from 30 to 60°C for all wells. These temperatures are the injection temperature at the reservoir perforation face and not surface injection temperature.

The extent of formation damage with respect to pour point temperature or adverse mobility effects considering cloud point temperature can be inferred from local grid refinement (10 m × 10 m) simulation around injection well J41-61 at perforation face injection temperature of 40°C

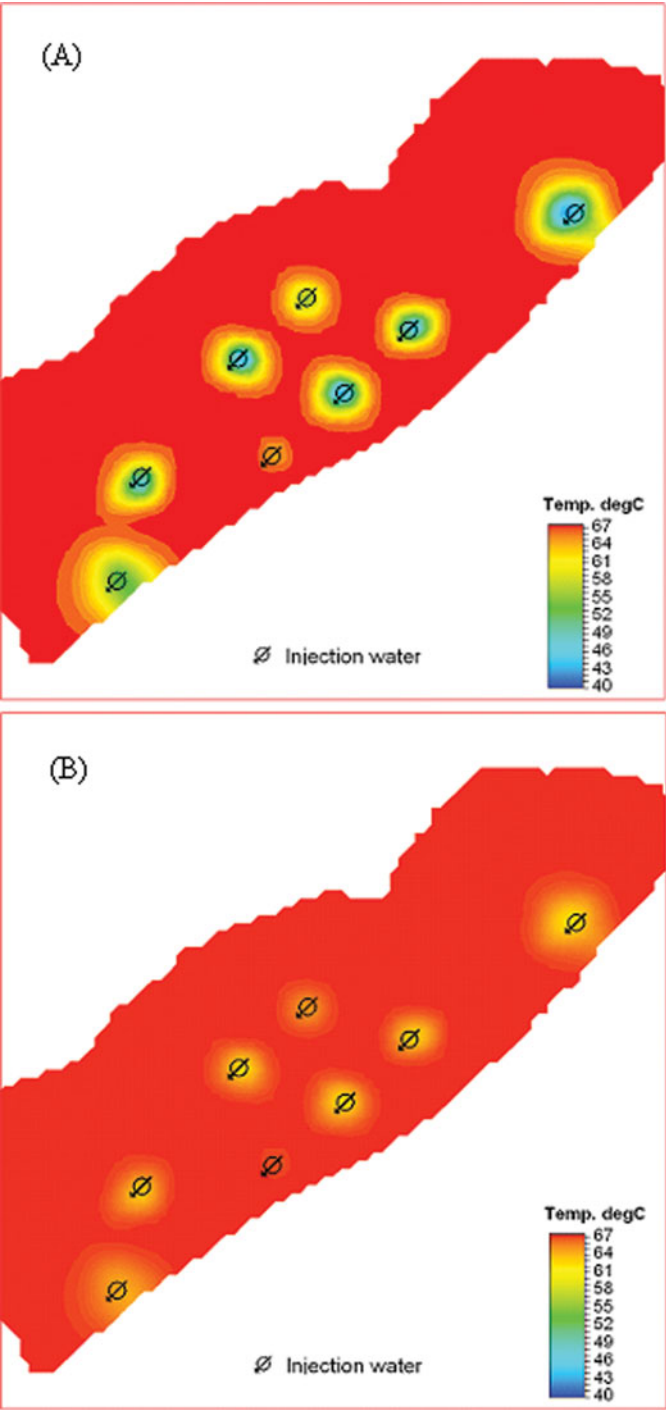


FIGURE 3 Reservoir performance history temperature distribution under the influence of nonisothermal water-flooding by 3/1993 the peak injection rate period for Zone 8: (A) injection temperature 30°C at perforation face and (B) injection temperature 60°C at perforation face.

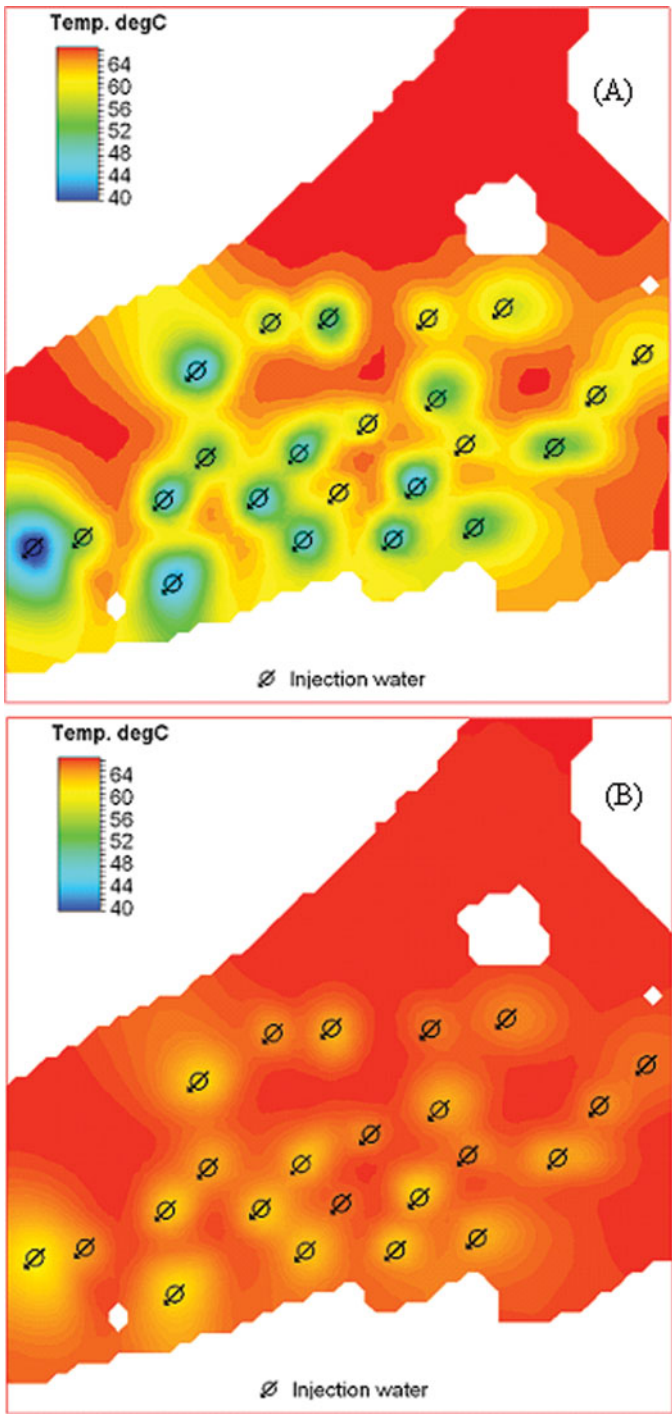


FIGURE 4 Reservoir performance history temperature distribution under the influence of nonisothermal water-flooding by 1/2002 the peak injection rate period for Zone 8: (A) injection temperature 30°C at perforation face and (B) injection temperature 60°C at perforation face.

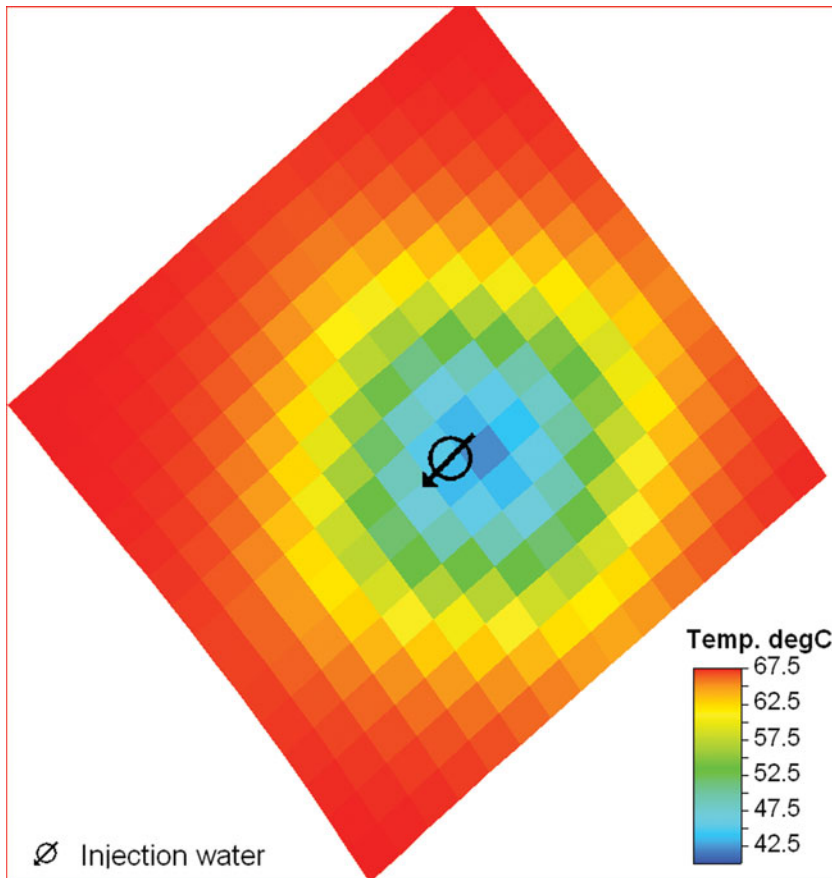


FIGURE 5 Temperature distribution grid block (10 m × 10 m) around injection well J41-61 at 40°C perforation face injection temperature for layer 8.

(Figure 5; corresponds to peak historical injection rate of 7,000 m³/month). The critical temperature window at peak injection rate for destabilizing in-situ fluid condition is 25–30 m from the injection well of study. The interesting outcome is that only during the high rate period is the vicinity of the well at high risk to damage. Damage as explained is with respect to the chosen and/or assumed no go temperature below 50°C based on fluid properties Table 1.

Optimal Injection Temperature

A temperature of about 50°C (average pour point temperature) is used to fix the optimal injection temperature at perforation zone in terms of reservoir temperature distribution. Optimal injection temperature is highly dependent on the injection rate. If injection temperature is to be optimized, one has to consider geothermal effect as well as the surface injection temperature to know the optimal temperature needed at the perforation face/zone. In light of this, surface injection temperature is computed in view of quantitatively estimating this temperature.

Ramey's (1962) approximate analytical solution for wellbore temperature profile was calibrated by four injection wells temperature profile. The optimal surface injection temperature based on the injection rate (160 m³/day) is 40°C. Temperature distribution due to injection temperature at the perforation faces (Figures 4a and 4b) for 30°C and 60°C shows that compared to the North Fault Block (Figures 3a and 3b) the influence of the non-isothermal injection on the South Fault Block has a far greater influence away from the injection well vicinity. This is certainly due to the higher injection rates, smaller flood pattern size, and reservoir continuity.

RESERVOIR MANAGEMENT STRATEGY

The way forward in optimizing recovery for the fault blocks is based on considering (a) flood pattern, (b) injection rate and temperature, (c) remaining oil distribution and saturation, (d) EOR schemes, and (e) the economics aspect. Taking into account the flood pattern, well spacing, and network, sedimentary microfacies and initial oil distribution, it is obvious that more recovery would be obtained from both blocks. In light of these, enhancing recovery is discussed with respect to various operational schemes based on non-isothermal flooding and economics of further exploitation.

North Fault Block

Four development schemes were investigated. Case 1 entails conversion of eight production wells into injection wells with no infill well to minimize cost. Focus was on the highly productive region with injection failure due to high initial non-isothermal injection rate associated with paraffin. Injection rate was fixed, similar to history performance away from cloud point temperature window. Flood pattern essentially converted to line drive. Case 2 is similar to Case 1 but with three infill wells, two of which are injection wells. An infill injection well to enhance flooding with four-spot inverted pattern. Inefficient flood pattern zone converted to five-spot inverted pattern and further exploiting poorly drained zones based on remaining oil and saturation distribution maps. Selective squeezing of injector wells due to oil saturation below 42%. Case 3 applied three horizontal wells to exploit unexplored regions of thin sand layers with operational scheme similar to base case. Case 4 is Case 1 and Case 3 put together. Cases 1, 2, and 4 narrowed producer-injector spacing to 200–300 m. The performance from these reduced flood networks are well observed in the predicted production rate based on history matched and calibrated numerical model (Figure 6).

Case 2 is more profitable as a choice to further develop the Block based on net present value and internal rate of return. However, Case 4 may be a better option due to pressure sustenance and the possibility of shutting off defaulting zones of high watercut.

South Fault Block

Attempt at forecasting reservoir performance and production beyond the history period gave unrealistic and higher production rates compared to the history end period with the history matched simulation model for the South Fault Block with matched oil initially in place. The unresolved high pressure makes the model unsuitable for further studies.

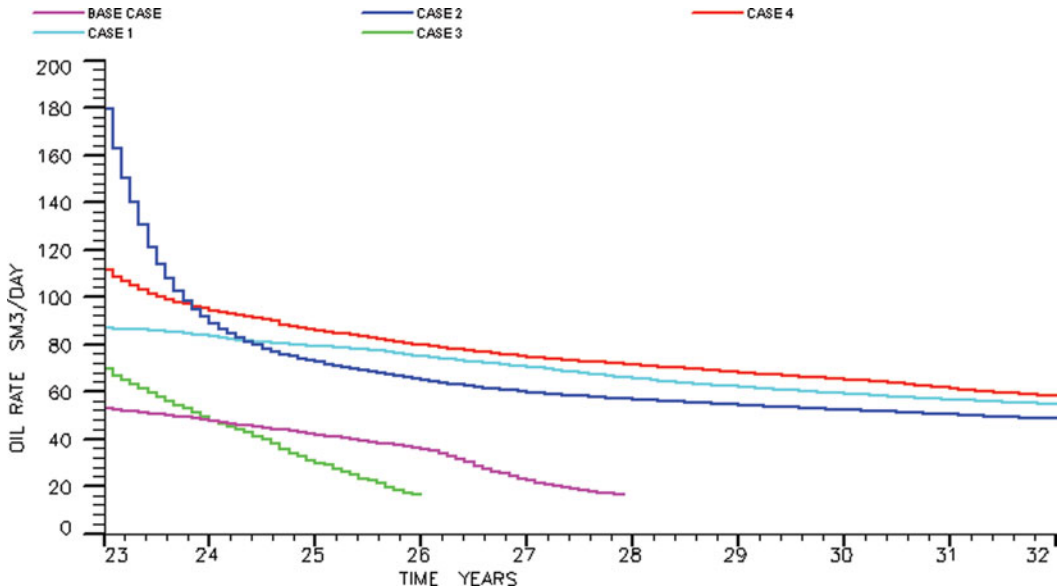


FIGURE 6 Production forecast of base case, Case 1, Case 2, Case 3, and Case 4.

CONCLUSION

High injection rates had presented injectivity issues compared to sustainable low rates due to avoiding both the cloud point and pour point window for the North and South Fault Blocks. Further cold waterflooding with low rates tied to perforation face temperature out of cloud point occurrence window enhanced production with optimal production and economics by reorientation of flooding network. For the South Fault Block, the inability to resolve high simulated pressure without adjusting history injection rates prevented the use of the model for prediction purposes to optimize recovery. Modeling the entire effect of paraffin crystallization, precipitation, and deposition on flow performance is challenging and complex. The effect of paraffin is confined to the near wellbore region under controlled injection for the case study with various schemes to enhance performance intermittently as required that are solely linked to the wellbore and its vicinity.

ACKNOWLEDGMENTS

The authors express their sincere gratitude for the support of this work by Liaohe Oilfield Company.

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